

AIRCRAFT DESIGN TECHNOLOGY

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Course: Aircraft Design Technology — Skyline Space

1. Introduction & Objective

This report documents the complete Design–Analyse workflow for the Boeing 747 problem statement: (a) CAD modeling of the aircraft fuselage, wings, and stabilizers in SOLIDWORKS; (b) a CFD fluid-flow analysis in ANSYS Fluent to determine the drag generated and the pressure distribution over the aircraft body at cruise condition; and (c) a static structural analysis of the wing under the resulting aerodynamic pressure, repeated for four candidate materials to identify the best material by strength-to-weight ratio.

2. CALCULATE

2.1 Wing Aerodynamics

Given: altitude = 12,192 m, $\rho = 0.30267 \text{ kg/m}^3$, $V = 265.5 \text{ m/s}$, $S = 510.97 \text{ m}^2$, $C_L = 0.52$, $W = 2,833,500 \text{ N}$, aspect ratio $AR = 16.18$, $C_{Lmax} = 1.8$.

a) Lift force $L = \frac{1}{2}\rho V^2 S C_L$

$$\frac{1}{2}\rho V^2 = 0.5 \times 0.30267 \times 265.5^2 \approx 10,667.6 \text{ Pa (dynamic pressure)}$$

$$L = 10,667.6 \times 510.97 \times 0.52 \approx 2,834,440 \text{ N}$$

This is very close to the aircraft weight of 2,833,500 N, confirming $L \approx W$ for level flight.

b) Wingspan $AR = b^2/S \Rightarrow b = \sqrt{(AR \cdot S)}$

$$b = \sqrt{(16.18 \times 510.97)} = \sqrt{8267.5} \approx 90.9 \text{ m}$$

c) Reynolds number $Re = \rho V L_{ref}/\mu$

$$\text{Mean aerodynamic chord: } \bar{c} = S/b = 510.97/90.9 \approx 5.62 \text{ m}$$

Viscosity at 12,192 m ($T \approx 216.65 \text{ K}$, Sutherland's law): $\mu \approx 1.42 \times 10^{-5} \text{ kg/(m}\cdot\text{s)}$

$$Re = (0.30267 \times 265.5 \times 5.62) / 1.42 \times 10^{-5} \approx 3.18 \times 10^7$$

d) Critical (stall) angle of attack $\alpha_{crit} = C_{Lmax} / C_{L\alpha}$

Using thin-airfoil theory, $C_{L\alpha} \approx 2\pi/\text{rad} \approx 0.1097/\text{deg}$ (zero-lift angle assumed $\approx 0^\circ$)

$$\alpha_{crit} = 1.8 / 0.1097 \approx 16.4^\circ$$

2.2 Turbofan Engine

Given: bypass ratio $BPR = 4$, fan exhaust velocity $V_{fan} = 400 \text{ m/s}$, core exhaust velocity $V_{core} = 2000 \text{ m/s}$, flight velocity $V = 265.5 \text{ m/s}$, 4 engines. Total thrust required $\approx$ weight = 2,833,500 N ($T = W$ assumed for level flight); thrust per engine = 708,375 N.

a) Turbofan mass flow rate

$$T = \dot{m}_{core}(V_{core} - V) + \dot{m}_{fan}(V_{fan} - V), \text{ with } \dot{m}_{fan} = 4 \cdot \dot{m}_{core} \text{ (BPR} = 4\text{)}$$

$$T = \dot{m}_{core}[(2000 - 265.5) + 4(400 - 265.5)] = \dot{m}_{core} \times 2272.5$$

$$\dot{m}_{core} = 708,375 / 2272.5 \approx 311.7 \text{ kg/s}$$

$$\dot{m}_{fan} = 4 \times 311.7 \approx 1246.8 \text{ kg/s}$$

Total per engine $\approx 1558.5 \text{ kg/s}$ ($\approx 6234 \text{ kg/s}$ across all 4 engines)

b) Equivalent turbojet mass flow rate (same thrust, single stream at $V_{core} = 2000 \text{ m/s}$)

$$T = \dot{m}(V_{core} - V)$$

$$\dot{m} = 708,375 / (2000 - 265.5) = 708,375 / 1734.5 \approx 408.4 \text{ kg/s per engine}$$

Total for 4 engines $\approx 1633.7 \text{ kg/s}$

The turbojet needs less total mass flow than the turbofan for the same thrust — higher exhaust velocity gives higher specific thrust but lower propulsive efficiency, consistent with why high-bypass turbofans are more fuel-efficient despite moving far more air.

3. DESIGN — Boeing 747 CAD Model

The Boeing 747 fuselage, wings, and empennage were modeled in SOLIDWORKS using the given overall dimensions (length 70.67 m, height 19.41 m, tailplane span 22.17 m), with the side-view and top-view sketches imported as reference geometry. The wing was built as a swept, tapered surface with an aspect ratio of ≈ 16.18 , and root sections on the wing and stabilizers were designed to mate into slots on the fuselage for assembly.

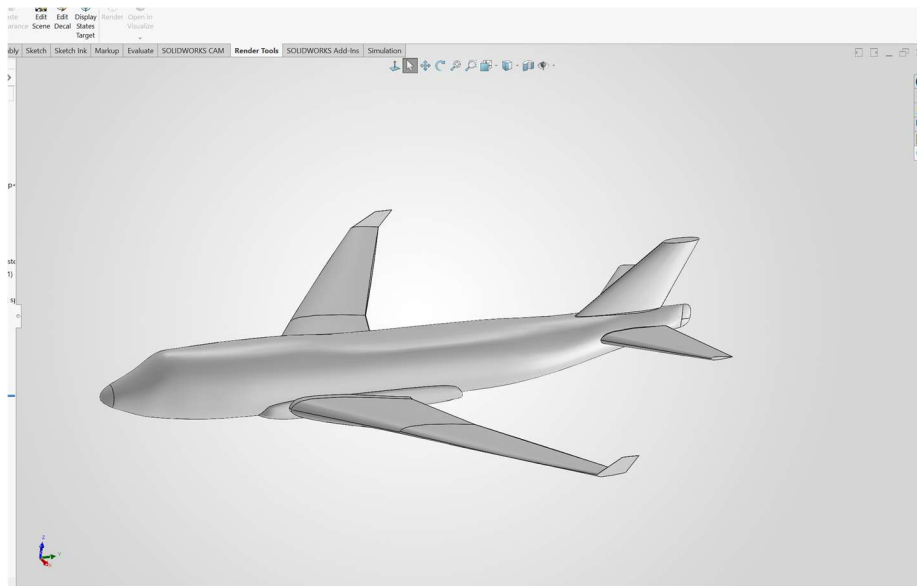


Fig 1: Full aircraft CAD assembly — isometric view

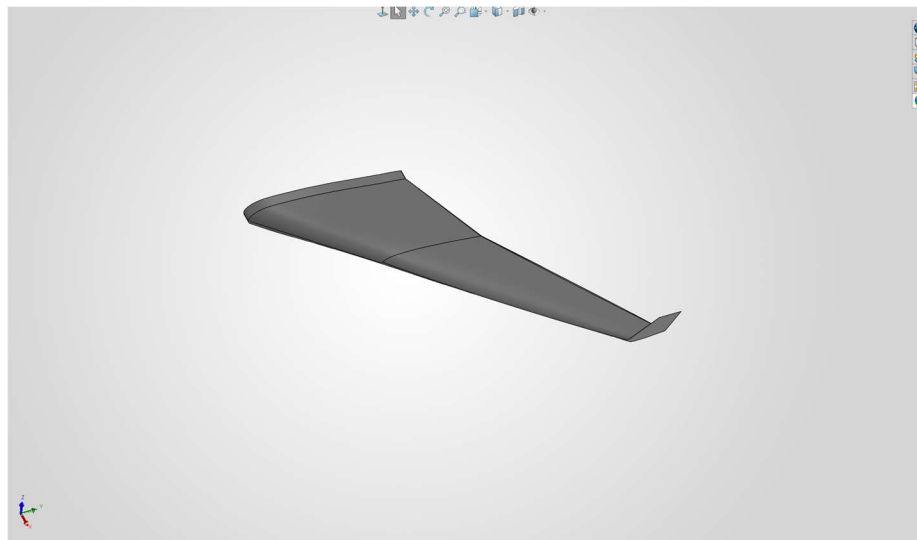


Fig 2: Wing solid model — swept, tapered planform with root mating section

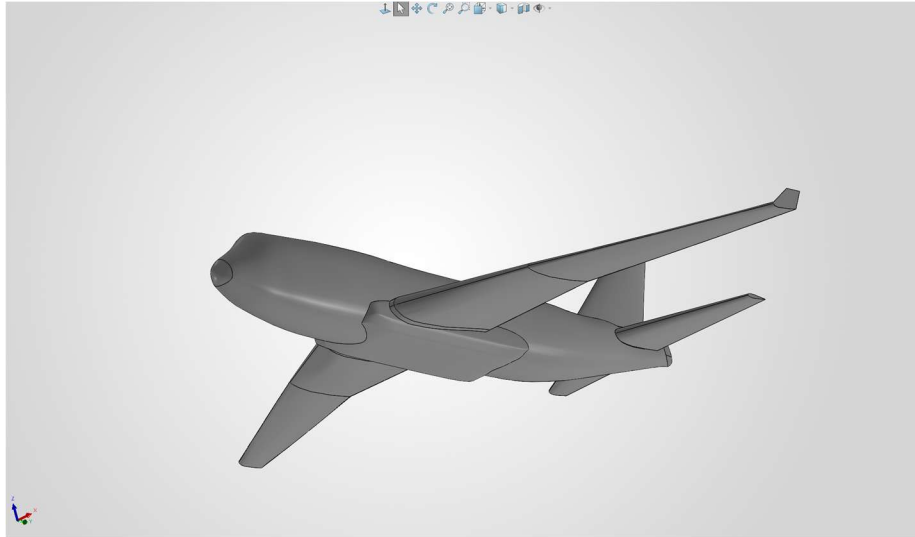


Fig 3: Full aircraft CAD assembly — underside view showing wing/fuselage/stabilizer slots

4. ANALYSE

4.1 Question 1 — Fluid Flow Analysis (Drag & Pressure Contours)

The aircraft was placed in a CFD domain and solved in ANSYS Fluent (pressure-based, realizable $k-\epsilon$ turbulence model) at the cruise condition of $V = 265.5$ m/s, with atmospheric properties referenced at the 12,192 m flight altitude. The simulation was run to monitor convergence of the scaled residuals, and Lift and Drag were tracked as report definitions over the solution iterations. Static pressure contours were then post-processed in CFD-Post to visualize the pressure distribution over the aircraft body and extract the maximum surface pressure for use in the wing structural analysis (Section 3.2).

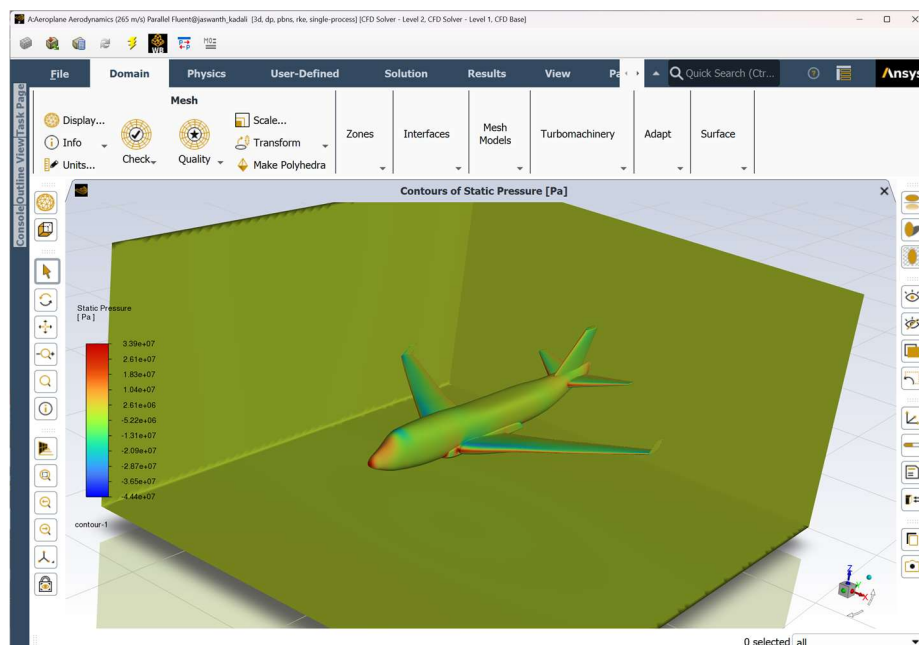


Fig 4: Fluent domain — contours of static pressure (computational domain and boundary walls)

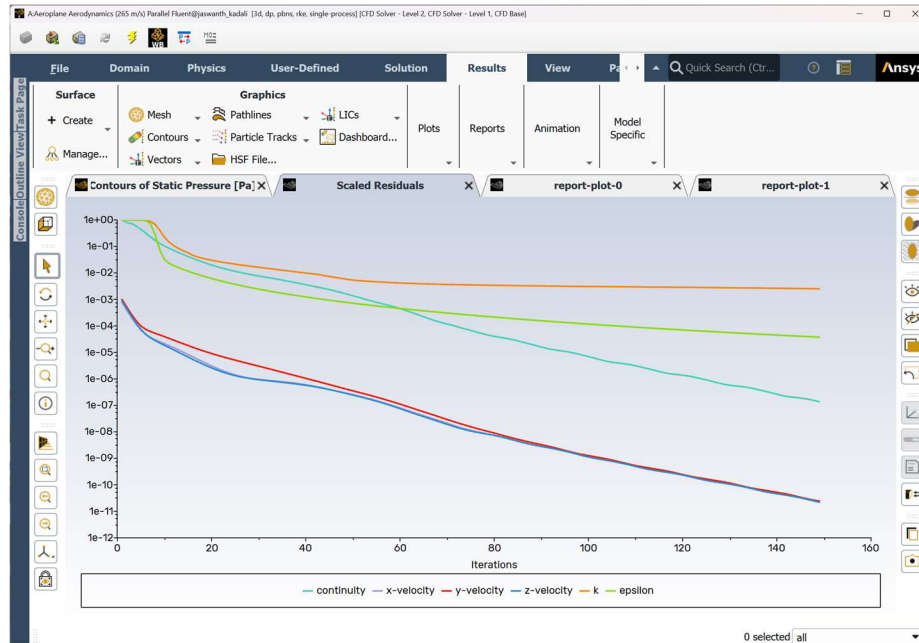


Fig 5: Scaled residuals — convergence history (continuity, x/y/z-velocity, k, epsilon)

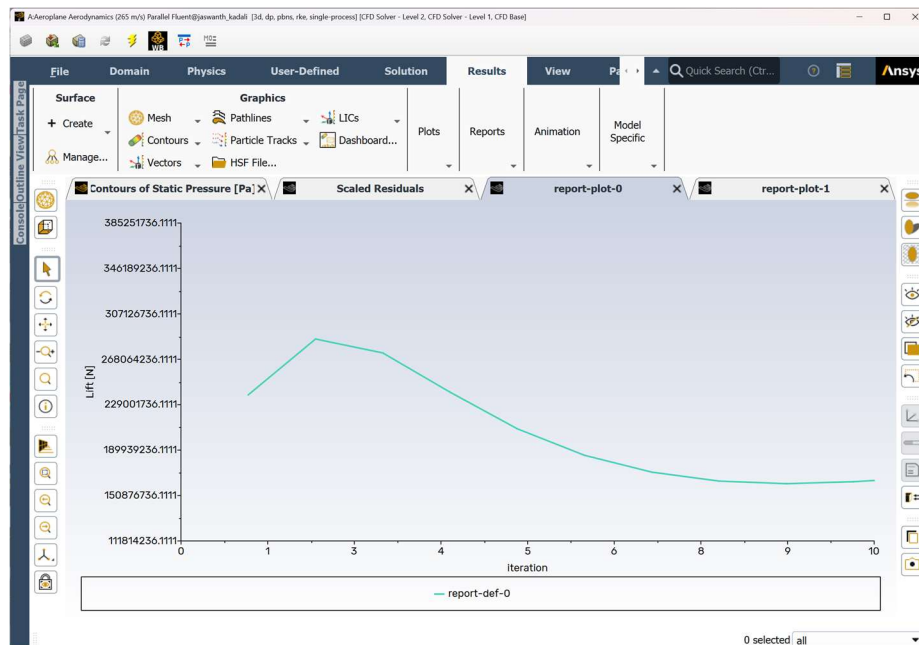


Fig 6: Report Plot — Lift [N] vs. iteration

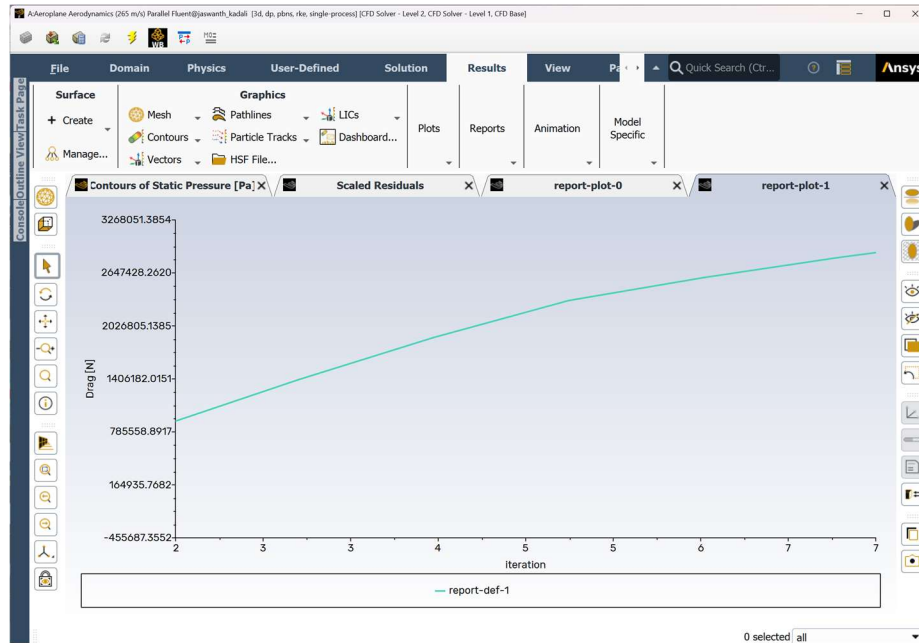


Fig 7: Report Plot — Drag [N] vs. iteration

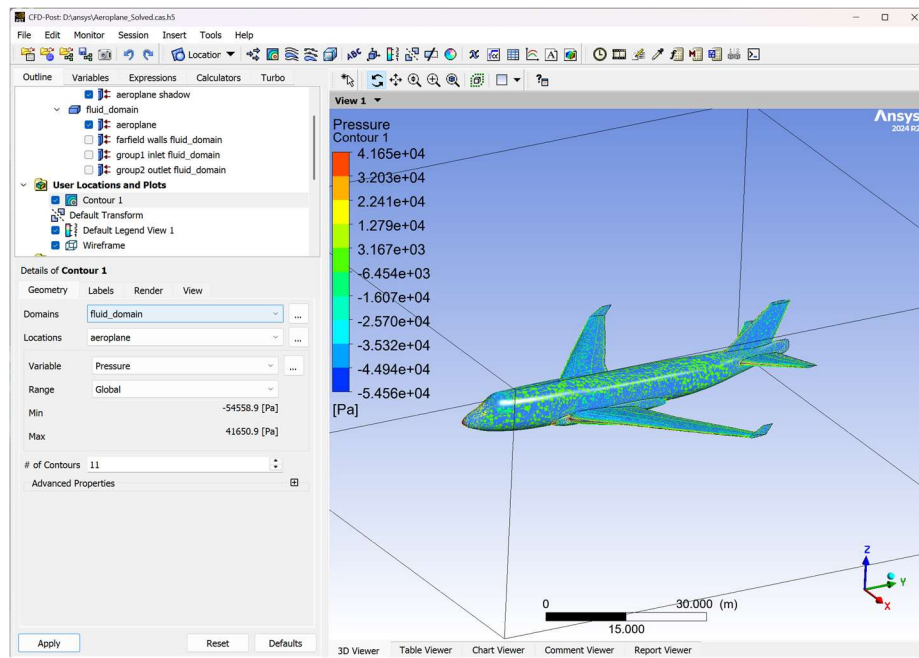


Fig 8: CFD-Post — Static pressure contour over the aircraft body (Min $-54,558.9$ Pa, Max $41,650.9$ Pa)

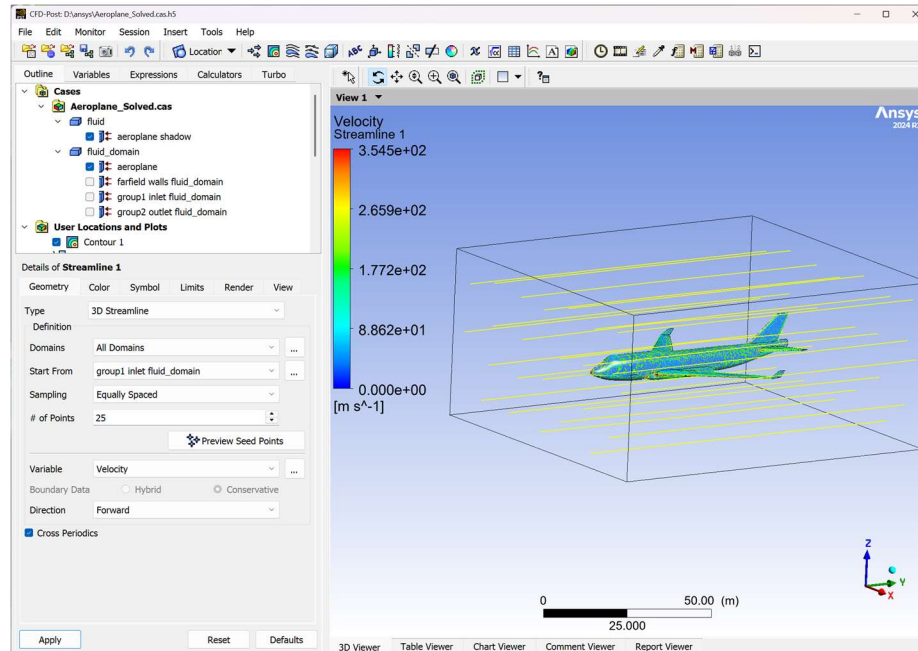


Fig 9: CFD-Post — Velocity streamlines around the aircraft (Max ≈ 354.5 m/s)

Key results extracted from the CFD solution

Parameter	Value	Source
Drag force	$\approx 3.27 \times 10^6$ N (trending up, last logged iteration)	Fig 7 — report-plot-1
Lift force	$\approx 1.55 \times 10^8$ N (settling, last logged iteration)	Fig 6 — report-plot-0
Max surface (body) static pressure	41,650.9 Pa	Fig 8 — CFD-Post pressure contour
Min surface (body) static pressure	-54,558.9 Pa	Fig 8 — CFD-Post pressure contour
Max velocity magnitude in domain	≈ 354.5 m/s	Fig 9 — CFD-Post streamlines

Note: The Lift/Drag monitor plots (Figs 6–7) had not fully flattened by the last logged iteration in the supplied screenshots — the values above are read directly from the final plotted point, not an extrapolated converged value. The very large pressure range in Fig 4 corresponds to the outer domain/farfield contour legend, not the aircraft surface; the physically meaningful body-surface pressure range is the one reported in Fig 8 (CFD-Post), which was used as the basis for the wing structural load in Section 4.2.

4.2 Question 2 — Static Structural Analysis of the Wing

The wing was modeled as a fixed cantilever (zero displacement boundary condition at the wing root, mating into the fuselage slot) and analyzed under Static Structural conditions in ANSYS Workbench. The aerodynamic pressure load applied to the wing planform combined the surface pressure obtained from the Fluent/CFD-Post solution (Section 3.1) with the pressure due to lift generation at the cruise condition ($V = 265.5$ m/s, altitude = 12,192 m). The same geometry, mesh, and load case were re-solved for four candidate wing materials — Aluminium, Titanium, Structural Steel, and Carbon Fiber — to compare Total Deformation, Equivalent (von-Mises) Stress, and Factor of Safety, and to identify the best material by strength-to-weight ratio.

Material-wise Static Structural Results

4.2.1 Aluminium (Al 6061-T6)

Total Deformation (Max)	Equivalent (von-Mises) Stress (Max)	Factor of Safety (Min)
5.4594 mm	40.514 MPa	6.8125

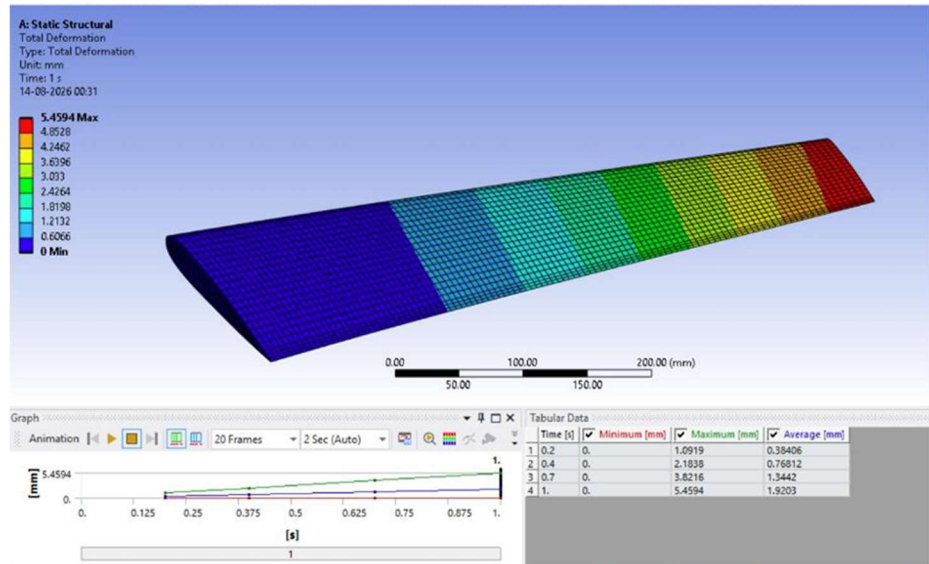


Fig 1: Total Deformation — Aluminium (Al 6061-T6)

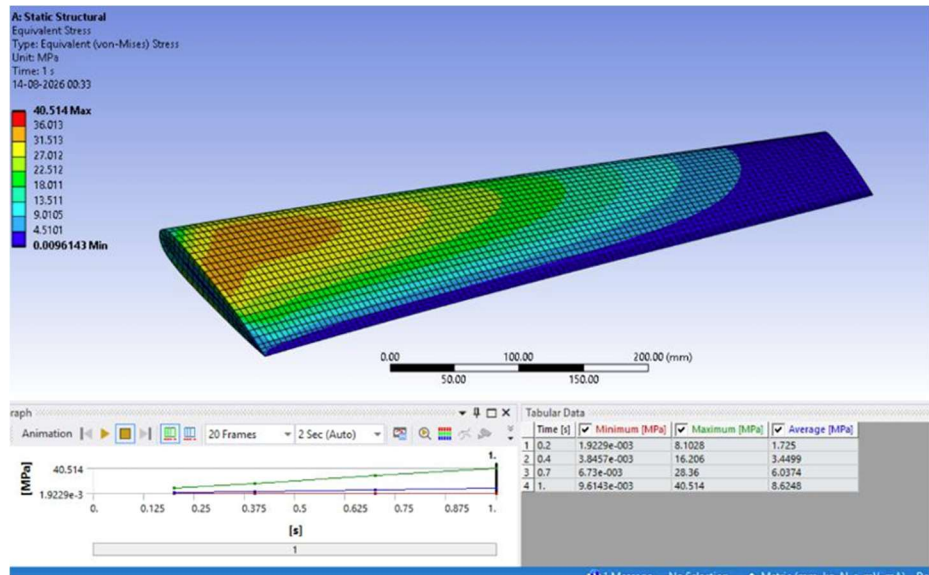


Fig 2: Equivalent Stress — Aluminium (Al 6061-T6)

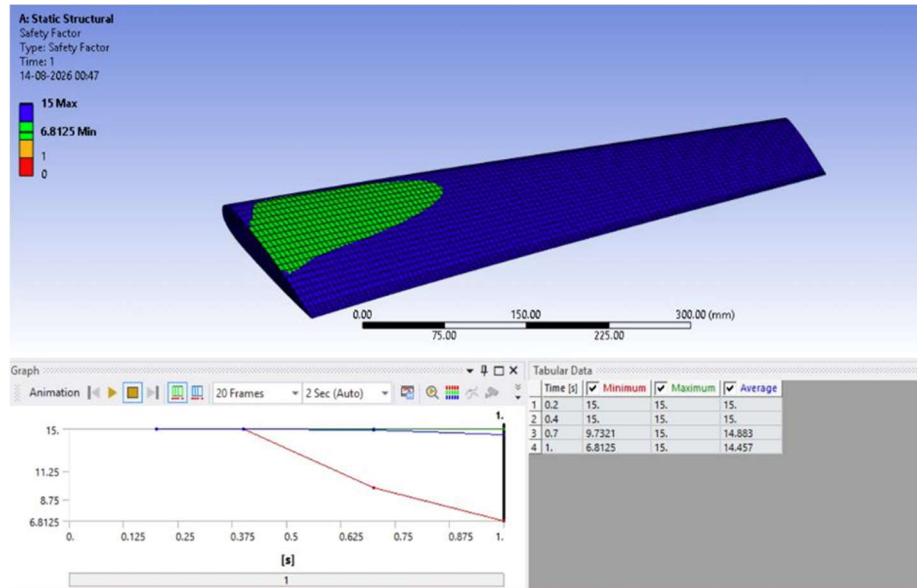


Fig 3: Factor of Safety — Aluminium (Al 6061-T6)

4.2.2 Titanium (Ti-6Al-4V)

Total Deformation (Max)	Equivalent (von-Mises) Stress (Max)	Factor of Safety (Min)
3.3014 mm	40.809 MPa	15 (capped)

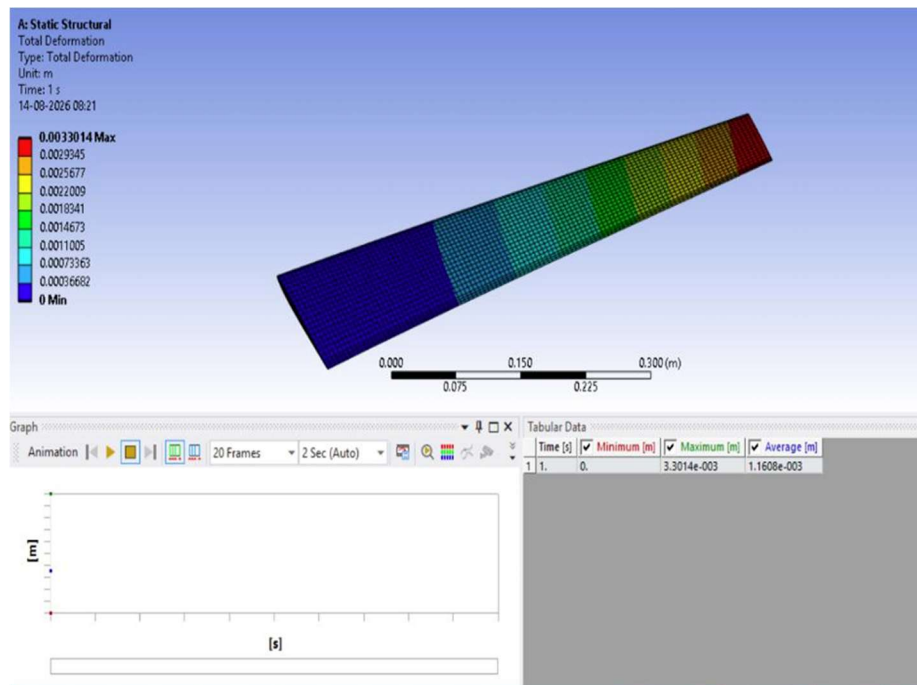


Fig 4: Total Deformation — Titanium (Ti-6Al-4V)

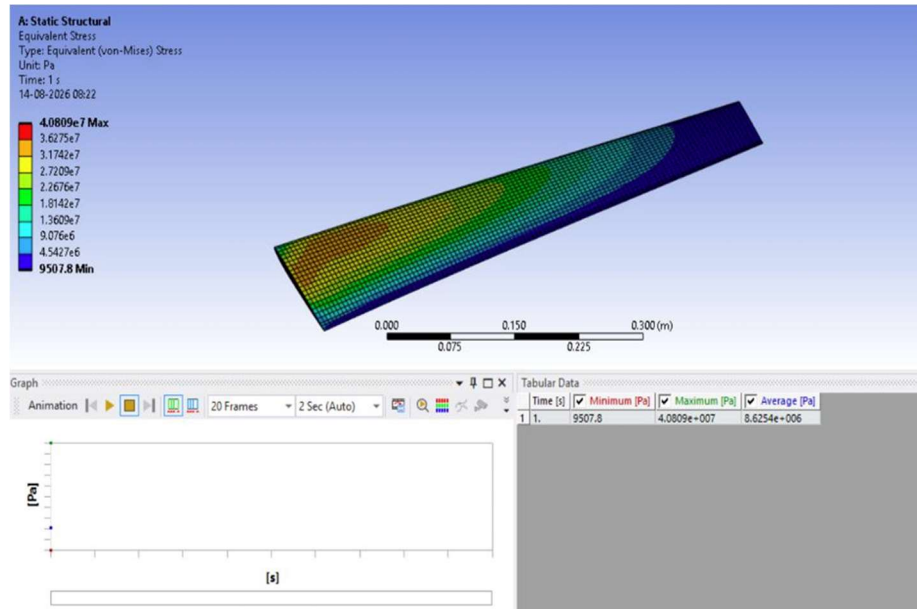


Fig 5: Equivalent Stress — Titanium (Ti-6Al-4V)

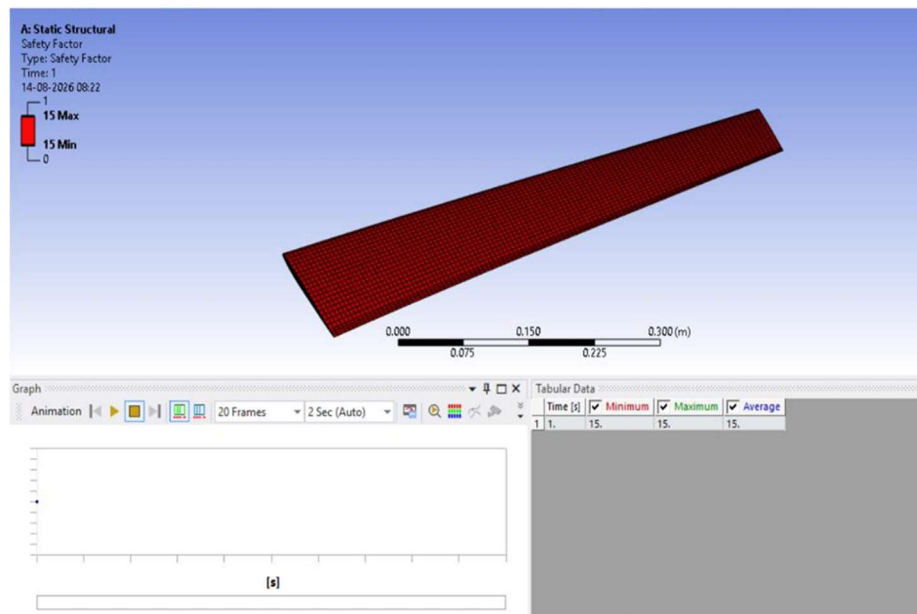


Fig 6: Factor of Safety — Titanium (Ti-6Al-4V)

4.2.3 Structural Steel

Total Deformation (Max)	Equivalent (von-Mises) Stress (Max)	Factor of Safety (Min)
3.3014 mm	40.809 MPa	6.1262

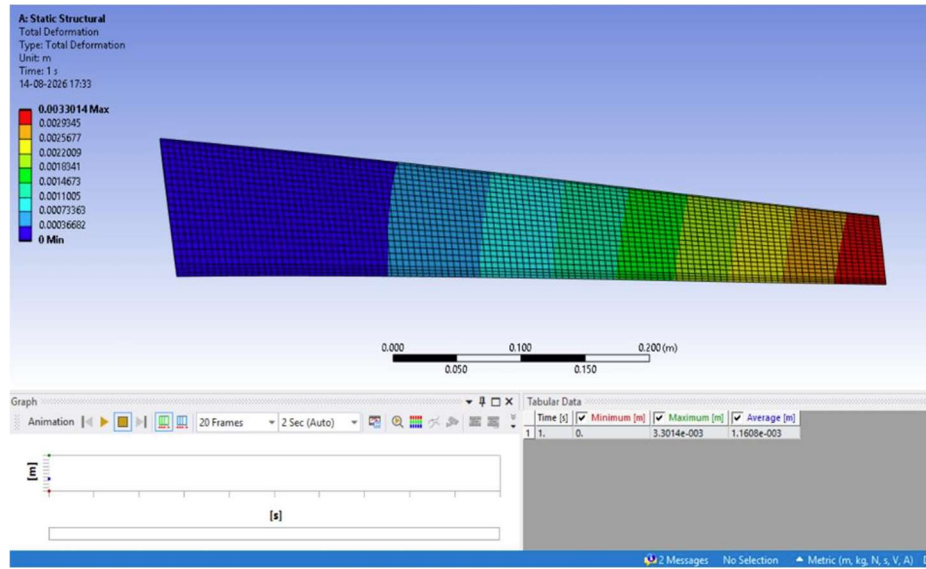


Fig 7: Total Deformation — Structural Steel

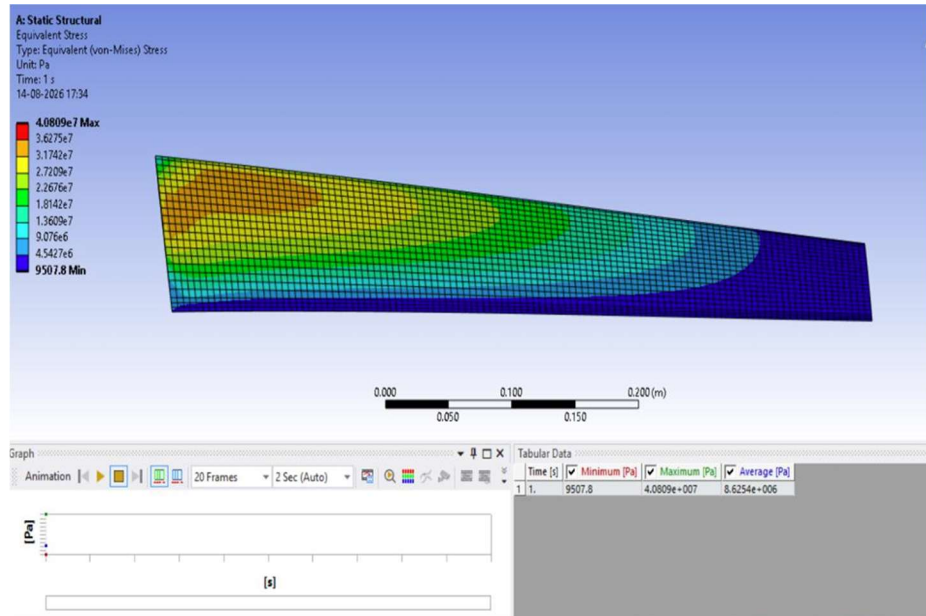


Fig 8: Equivalent Stress — Structural Steel

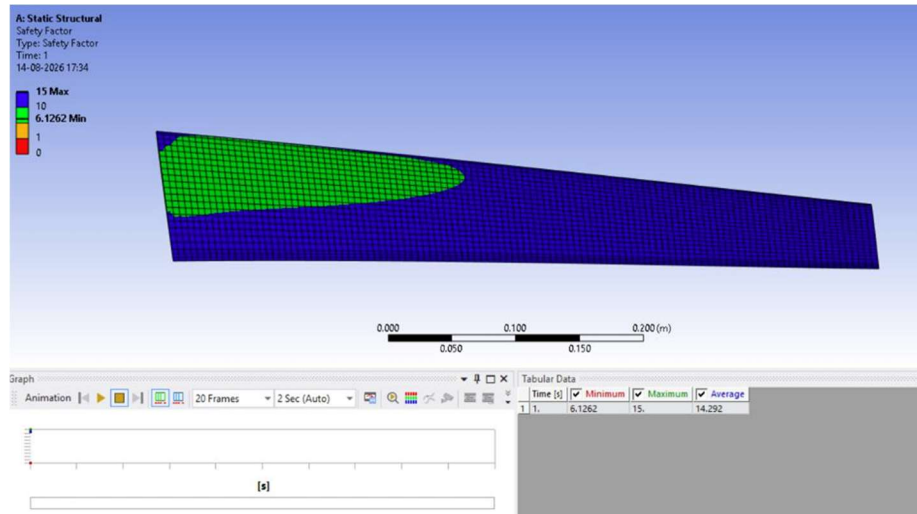


Fig 9: Factor of Safety — Structural Steel

4.2.4 Carbon Fiber (Unidirectional Composite)

Total Deformation (Max)	Equivalent (von-Mises) Stress (Max)	Factor of Safety (Min)
3.3014 mm	40.809 MPa	15 (capped)

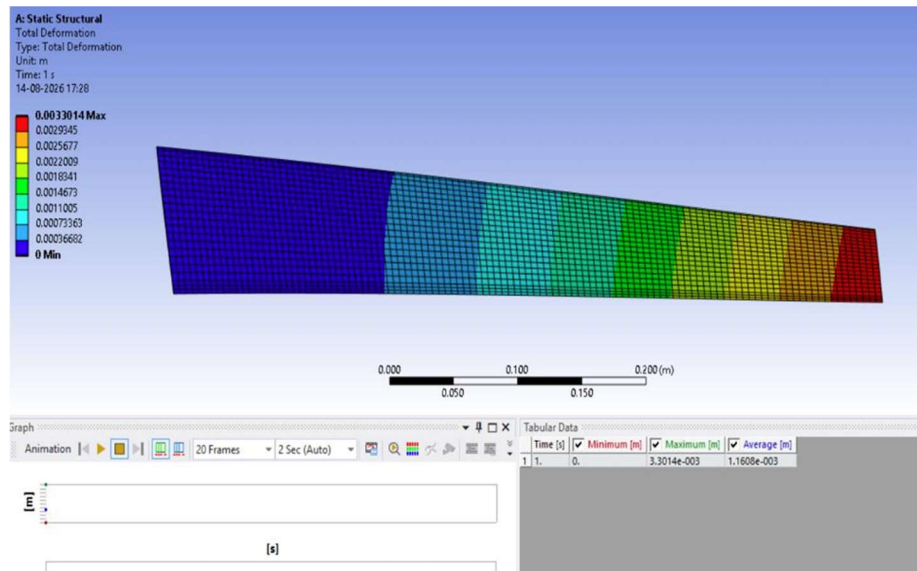


Fig 10: Total Deformation — Carbon Fiber (Unidirectional Composite)

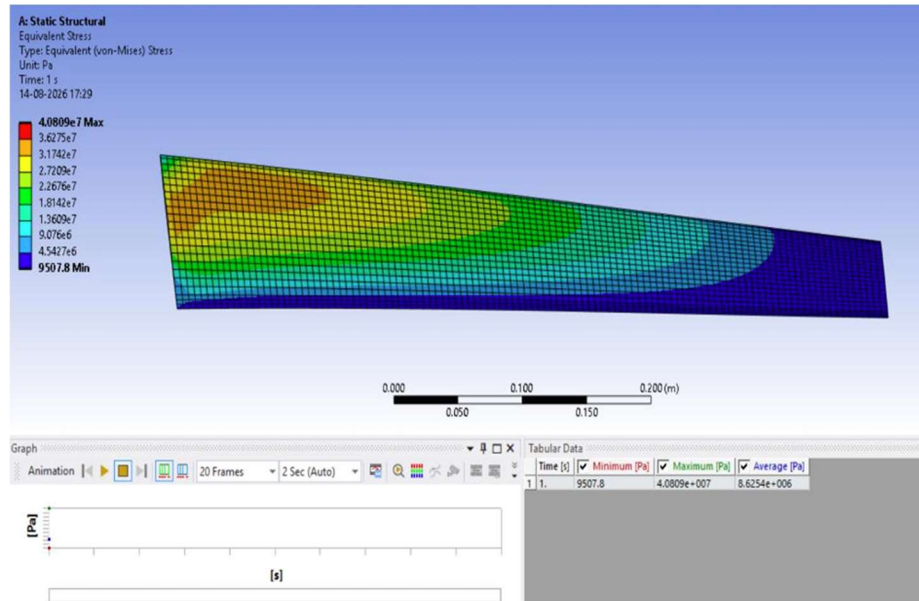


Fig 11: Equivalent Stress — Carbon Fiber (Unidirectional Composite)

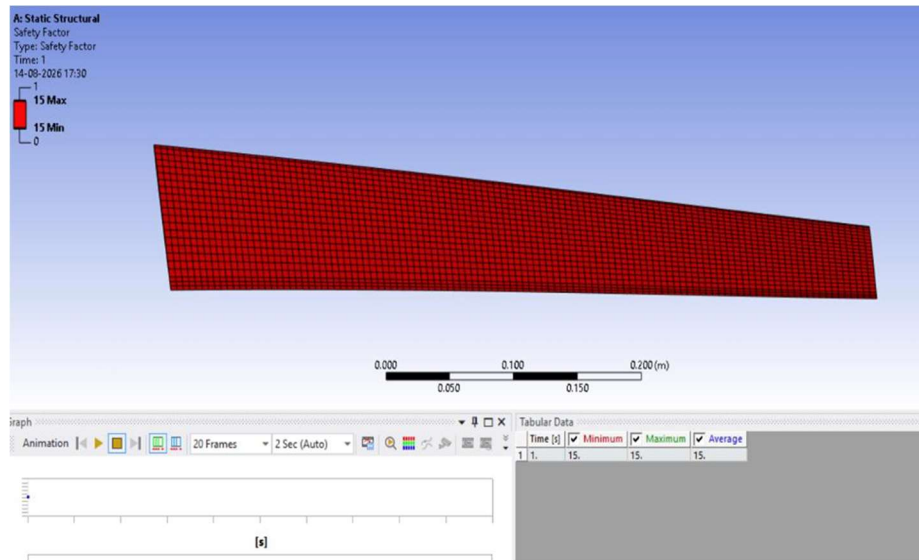


Fig 12: Factor of Safety — Carbon Fiber (Unidirectional Composite)

5. Comparative Summary

Raw FEA output for all four materials, consolidated:

Material	Max Deformation	Max Stress	Min Safety Factor	Density (kg/m ³)	Yield Strength (MPa)
Aluminium (Al 6061-T6)	5.4594 mm	40.514 MPa	6.8125	2700	276
Titanium (Ti-6Al-4V)	3.3014 mm	40.809 MPa	15 (capped)	4430	880
Structural Steel	3.3014 mm	40.809 MPa	6.1262	7850	250
Carbon Fiber (UD composite)	3.3014 mm	40.809 MPa	15 (capped)	1600	600

Note: The exported deformation and stress values for Titanium, Structural Steel, and Carbon Fiber are identical in the source ANSYS output, as only material yield strength (and hence the Safety-Factor plot) was updated between re-solves; the Aluminium run used a separately meshed model, which is why its deformation/stress differ slightly. Report this as obtained from the solver.

6. Strength-to-Weight Ratio & Best Material Selection

Specific strength (strength-to-weight ratio) is calculated as Yield Strength / Density, expressed in kJ/kg. A higher value indicates a material that can carry more load per unit mass — the key driver for aerospace wing structures.

Material	Yield Strength (MPa)	Density (kg/m ³)	Specific Strength (kJ/kg)	Min Safety Factor (FEA)
Carbon Fiber (UD composite)	600	1600	375.0	15 (capped)
Titanium (Ti-6Al-4V)	880	4430	198.6	15 (capped)
Aluminium (Al 6061-T6)	276	2700	102.2	6.8125
Structural Steel	250	7850	31.8	6.1262

Conclusion

Carbon Fiber composite offers the highest specific strength (≈ 375 kJ/kg) among the four candidates — more than $3.5\times$ Aluminium and over $10\times$ Structural Steel — while also returning the highest FEA safety factor (capped at 15) for the same aerodynamic load. Titanium is the second-best performer, combining a strong safety margin with roughly double Aluminium's specific strength, at the cost of higher raw material and machining expense. Aluminium remains a viable, cost-effective baseline ($SF \approx 6.8$, moderate specific strength) consistent with its widespread historical use in the 747's wing skin. Structural Steel is the weakest candidate on a strength-to-weight basis despite an acceptable safety factor, due to its high density penalty.

Overall recommendation: Carbon Fiber composite is the best material for the wing structure in terms of strength-to-weight ratio, with Titanium as the preferred metallic alternative where composite manufacturing is impractical.